CYBER SECURITY INTERNSHIP

TASK 4

Setup and Use a Firewall on Linux Using UFW

Practical Configuration, Rule Validation and Testing Report

Platform	Kali Linux (firewall target) and Windows 10 (test client)
Firewall utility	UFW – Uncomplicated Firewall
Test network	VMware virtual network
Kali IP address	192.168.182.128
Windows IP address	192.168.182.129
Tested service	TCP port 23 (Telnet)

Prepared as practical evidence for the internship task. Screenshots in this report are taken from the user's Kali Linux and Windows 10 virtual machines.

1. Objective

The objective of this practical was to configure and test basic firewall rules that control inbound and outbound network traffic. UFW was used on Kali Linux to allow SSH traffic on TCP port 22 and deny inbound Telnet traffic on TCP port 23.

2. Lab Environment

Component	Configuration
Firewall host	Kali Linux
Firewall tool	UFW
Testing client	Windows 10 x64 virtual machine
Virtualization	VMware Workstation
Kali IPv4 address	192.168.182.128
Windows IPv4 address	192.168.182.129
Firewall rule under test	Deny inbound TCP/23

3. Firewall Configuration

The following commands were used to install UFW, define the default traffic policies, add the SSH allow rule, add the Telnet deny rule, and enable the firewall.

```
sudo apt update
sudo apt install ufw -y
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp
sudo ufw deny 23/tcp
sudo ufw enable
sudo ufw status verbose
```

The active configuration showed the default policy as deny incoming and allow outgoing. The rule list showed TCP/22 as ALLOW IN and TCP/23 as DENY IN. UFW also displayed corresponding IPv6 rules.

4. Initial Firewall Evidence

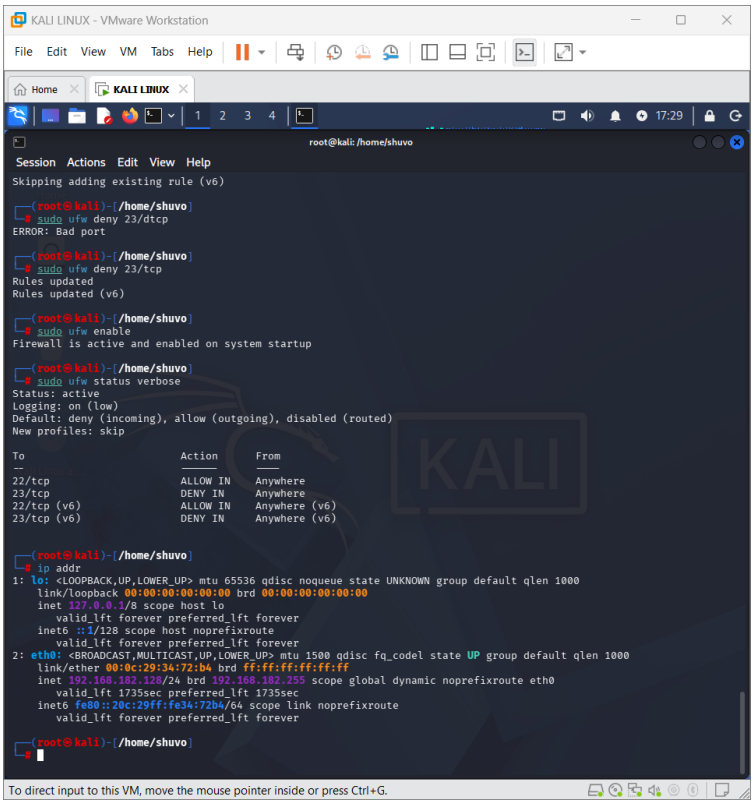


Figure 1. Initial Kali Linux UFW configuration and interface information. The screenshot shows UFW active, inbound traffic denied by default, outbound traffic allowed, TCP/22 allowed and TCP/23 denied.

5. Firewall Testing Procedure

A temporary TCP listener was started on the Kali Linux host on port 23. The Windows 10 virtual machine was then used as a separate client to test connectivity to the Kali host.

On Kali Linux (target):

```
nc -lvnp 23
```

On Windows PowerShell (client):

```
Test-NetConnection 192.168.182.128 -Port 23
```

6. Test Results

The Windows PowerShell output reported PingSucceeded as True and TcpTestSucceeded as False. This means the Kali host was reachable over the network, but a TCP connection to port 23 could not be established during the test. The Windows source address was 192.168.182.129 and the remote Kali address was 192.168.182.128.

Test item	Observed result	Interpretation
Ping to Kali	True	Host was reachable
TCP connection to port 23	False	TCP connection was not established
Windows source IP	192.168.182.129	Testing client
Kali destination IP	192.168.182.128	Firewall host

7. Screenshot Evidence – Kali Linux Listener

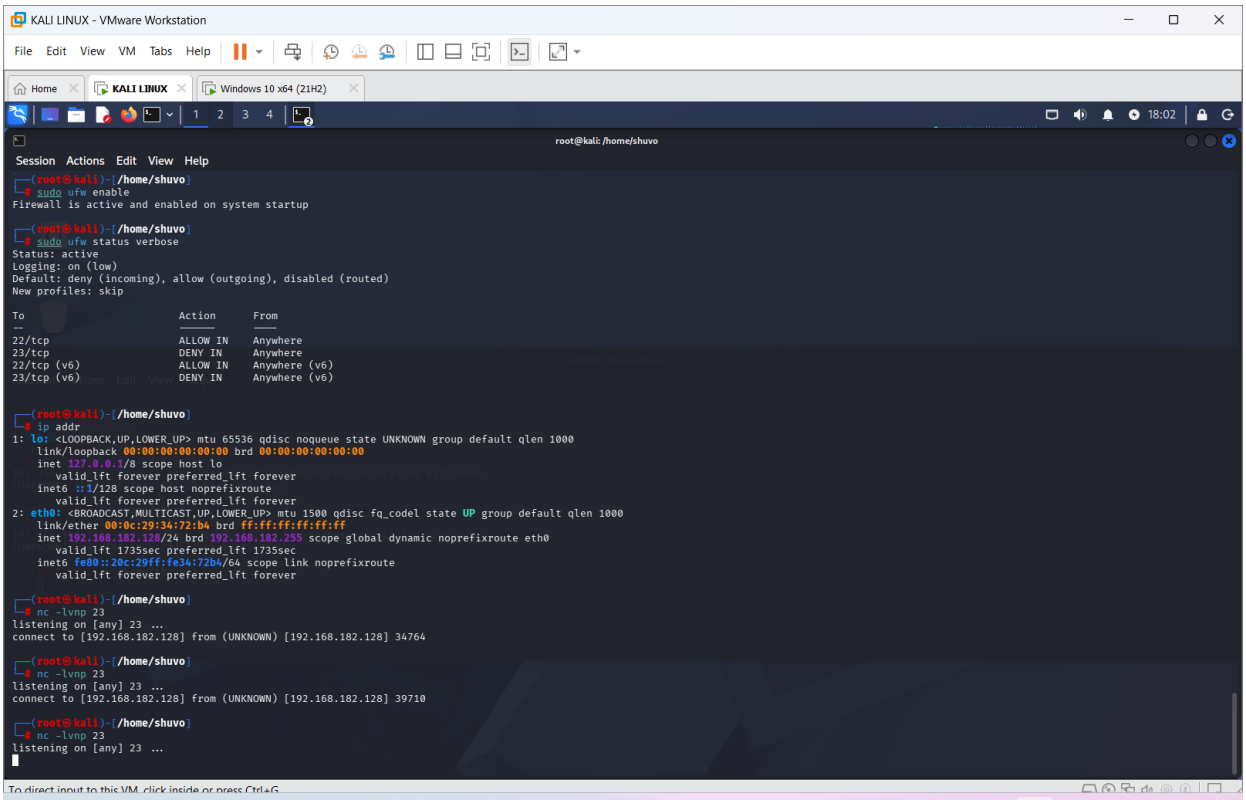


Figure 2. Kali Linux terminal showing the UFW rules and a Netcat listener bound to TCP port 23 during the external test.

8. Screenshot Evidence – Windows 10 Test

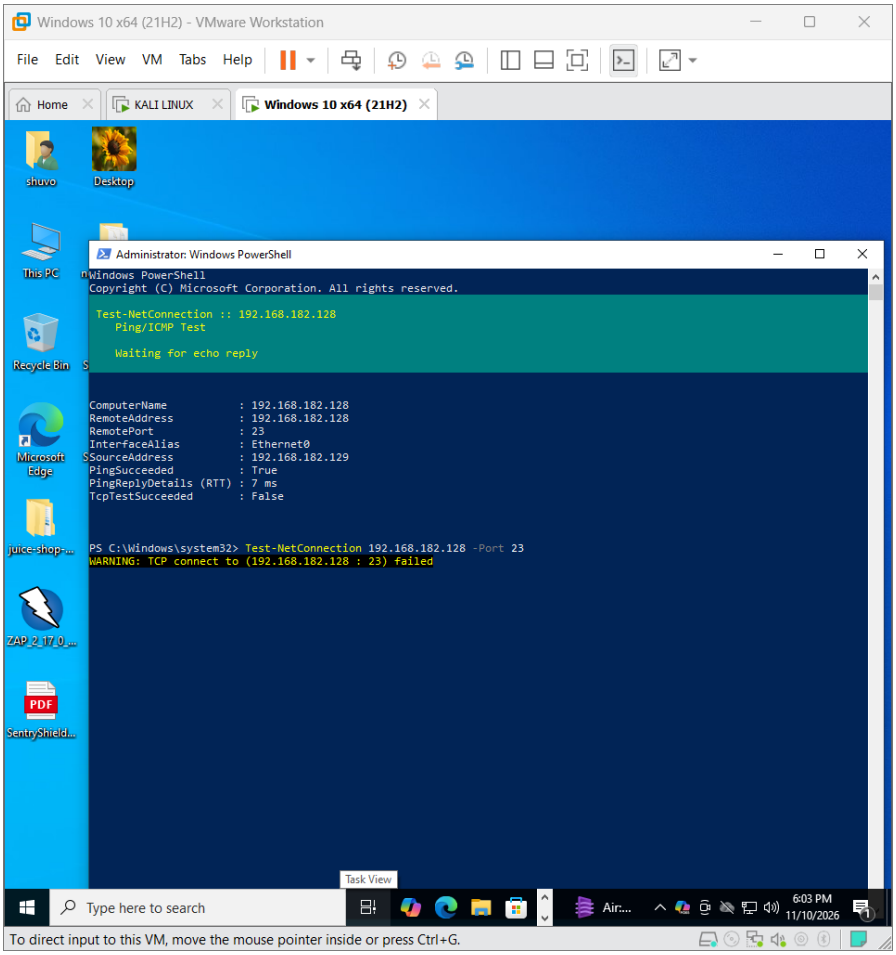


Figure 3. Windows PowerShell test. PingSucceeded is True, while TcpTestSucceeded is False for remote TCP port 23 on 192.168.182.128.

Interpretation note: The failed TCP connection is consistent with the configured UFW deny rule. A failed TCP test by itself does not prove which device or policy caused the failure; in this lab, the Kali UFW rule was active and the Windows client could successfully ping the Kali host. The listener was running on Kali during the test.

9. Cleanup and Reset

After the evidence was captured, the temporary listener should be stopped with Ctrl+C. The temporary Telnet deny rule can be removed with:

```
sudo ufw delete deny 23/tcp
sudo ufw status numbered
```

To reset UFW to its default reset state and remove the custom UFW rules:

```
sudo ufw reset
# Confirm with y when prompted
sudo ufw status verbose
```

After a successful reset, UFW should report Status: inactive. Note that ufw reset restores UFW's defaults; it does not necessarily restore every firewall setting managed outside UFW or reproduce the exact configuration that existed before the lab.

10. Key Concepts

Firewall: A security control that filters network traffic according to configured rules.

Inbound rule: Controls connections entering the protected host.

Outbound rule: Controls connections initiated from the protected host toward other systems.

Stateful firewall: Tracks connection state and uses that context when evaluating packets. A stateless firewall evaluates packets independently without maintaining connection state.

UFW: A simplified command-line interface for managing Linux firewall rules.

Telnet (TCP/23): A legacy remote terminal protocol that does not encrypt its traffic by default. SSH is generally preferred for secure remote administration.

NAT: Network Address Translation modifies IP address information as traffic passes through a router or firewall, commonly allowing private hosts to share a public address.

11. Conclusion

UFW was installed and configured on Kali Linux. The default policy was set to deny incoming traffic and allow outgoing traffic. SSH on TCP port 22 was allowed, and inbound Telnet traffic on TCP port 23 was denied. Testing from a separate Windows 10 virtual machine showed that the Kali host responded to ping while the TCP connection test to port 23 failed. This practical demonstrated basic firewall administration, rule configuration, and cross-machine connectivity testing.

12. Evidence and Submission Checklist

Deliverable	Status
UFW configuration and rule listing screenshot	Included
Kali listener screenshot	Included
Windows PowerShell test screenshot	Included
Commands used	Documented
Cleanup/reset commands	Documented
GitHub repository link	To be added by student

Submission note: Add this PDF and the original screenshots to a dedicated GitHub repository for Task 4. Add the repository URL to your internship submission form.

Appendix A – Interview Questions and Answers

The following answers address the eight interview questions listed in the Task 4 assignment brief. They are written for revision and practical discussion.

Q1. What is a firewall?

A firewall monitors and controls incoming and outgoing network traffic according to predefined security rules. It acts as a protective barrier between trusted and untrusted networks and helps prevent unauthorized access and unwanted communication.

Example: UFW can deny incoming TCP traffic on port 23 while allowing SSH on port 22.

Q2. What is the difference between a stateful and stateless firewall?

Stateful firewall: Tracks active network connections and uses connection state when making filtering decisions.
Stateless firewall: Examines packets independently against configured rules and does not maintain connection state. It generally makes decisions using packet attributes such as source, destination, protocol and port.

Q3. What are inbound and outbound rules?

Inbound rules control traffic entering a computer or network. **Outbound rules** control traffic leaving a computer or network.

Inbound example: `sudo ufw deny 23/tcp`
Outbound example: `sudo ufw deny out 25/tcp`

Q4. How does UFW simplify firewall management?

UFW (Uncomplicated Firewall) provides a straightforward command-line interface for managing Linux firewall rules. It allows administrators to enable or disable the firewall, allow or deny ports, manage IP-based rules and inspect the configuration without manually writing low-level firewall rules.

Examples:

```
sudo ufw enable
sudo ufw allow 22/tcp
sudo ufw deny 23/tcp
sudo ufw status verbose
```

Q5. Why should we block port 23 (Telnet)?

TCP port 23 is commonly used by Telnet. Telnet does not encrypt communication by default, so usernames, passwords and other data may be exposed to someone monitoring the network. Blocking unnecessary Telnet access reduces this risk. SSH, typically using TCP port 22, is generally preferred for encrypted remote administration.

Q6. What are common firewall configuration mistakes?

Common mistakes include allowing unnecessary ports, applying incorrect rules, failing to restrict untrusted sources, confusing inbound and outbound policies, overlooking IPv6 rules, failing to review logs, and not testing changes. On remote systems, administrators can also accidentally lock themselves out by enabling a firewall before allowing the required management connection.

Q7. How does a firewall improve network security?

A firewall enforces access-control policies by allowing or blocking traffic according to configured criteria. It can restrict exposed services and ports, limit access from particular IP addresses, reduce the system's attack surface and provide logs that help administrators investigate unexpected traffic. It is one layer of defense and should be combined with patching, authentication and endpoint security.

Q8. What is NAT in firewalls?

NAT (Network Address Translation) changes IP address information in packet headers as traffic passes through a router or firewall. It is commonly used to let multiple devices with private IP addresses share a public IPv4 address.

Types:
Static NAT: Fixed one-to-one address mapping.
Dynamic NAT: Maps private addresses to available public addresses from a pool.
PAT (Port Address Translation): Allows multiple private devices to share one public IP address by distinguishing sessions using port numbers.

Practical example from this lab: Kali Linux was configured with UFW to allow inbound SSH on TCP/22 and deny inbound Telnet on TCP/23. A Windows 10 VM was used to test connectivity. In a report, distinguish between a rule being present and successful enforcement being verified.